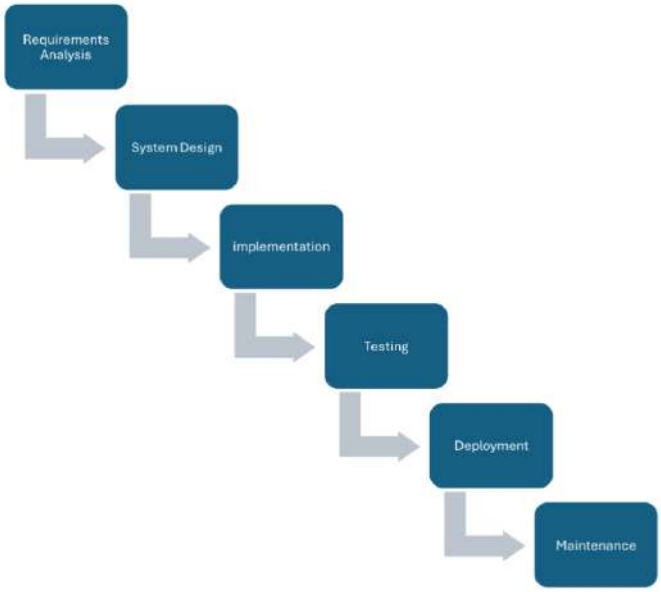
DAY 1
Activity 1
Software Development Life Cycle (SDLC)

The SDLC is a **structured process** used by software teams to design, develop, test, and deliver high-quality software. Below are the main phases represented in the flowchart.



SDLC Phases

1. Requirements Gathering & Analysis

- **Goal:** Understand what the customer or business needs.

Activities include:

- Conducting meetings with stakeholders
- Documenting functional & non-functional requirements
- Feasibility analysis (technical, financial, operational)

2. System Design

- **Goal:** Create the architecture and blueprint of the system.

Activities include:

- Designing UI/UX
- Database design
- System architecture diagrams

3. Implementation / Development

- **Goal:** Developers write the actual code.

Activities include:

- Coding based on design

- Following coding standards
- Unit testing by developers

4. Testing

- **Goal:** Identify and fix defects before releasing the product.

Activities include:

- Integration testing
- System testing
- UAT (User Acceptance Testing)
- Bug fixing

5. Deployment

- **Goal:** Release the finished product to production.

Activities include:

- Deploying to servers/app stores
- Release management
- User training

6. Maintenance

- **Goal:** Keep the software updated and functional after release.

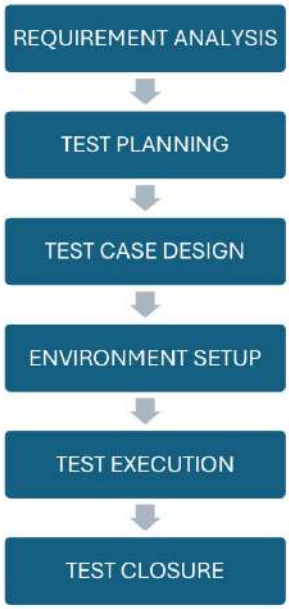
Activities include:

- Fixing post-deployment bugs
- Enhancements & new features
- Performance optimization
- Security updates

Activity 2
STLC Software Testing Life Cycle

STLC (Software Testing Life Cycle) is a sequence of steps followed during the testing o software.
Just like SDLC focuses on software development, STLC focuses only on the testing pro
It ensures:

- Quality of the product
- Early detection of defects
- Systematic and organized testing process



1. Requirement Analysis: Testers study project requirements to understand what nee tested.
Activities:

- Analyze functional & non-functional requirements
- Identify testable and non-testable requirements
- Identify testing priorities and risk.

2. Test Planning: A strategy is created for how testing will be done.

Activities:

- Define testing scope

- Prepare test plan
- Estimate effort and resources
- Identify tools (JIRA, Selenium, TestRail)

3. Test Case Design: Test cases are written based on requirements.

Activities:

- Create test scenarios
- Write test cases
- Prepare test data
- Review and approve test cases

4. Environment Setup: Test environment is prepared for test execution.

Activities:

- Set up QA environment
- Install builds, databases, tools
- Validate environment readiness
- Coordinate with devOps team

5. Test Execution: Actual testing happens here.

Activities:

- Execute test cases
- Log defects in bug tracking tools
- Retest & regression testing
- Track defect life cycle

6. Test Closure: Final stage of STLC after completing testing.

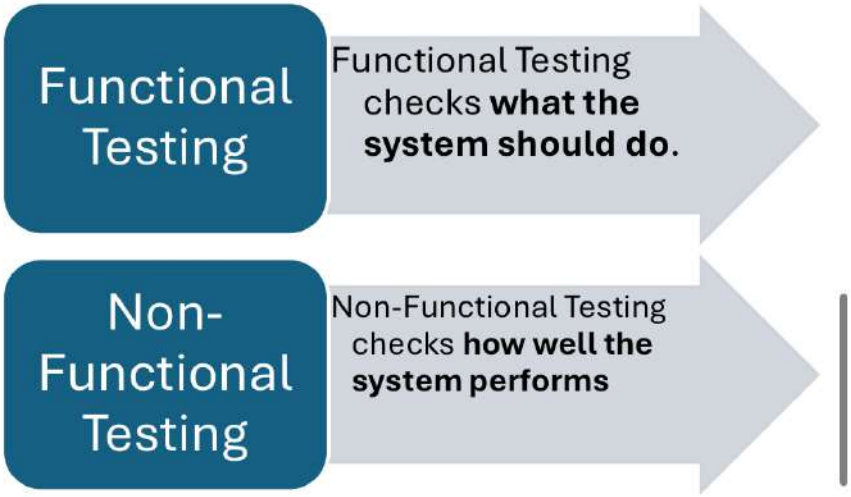
Activities:

- Prepare Test Summary Report
- Analyze testing metrics
- Document lessons learned
- Sign-off from QA team

ACTIVITY 3

TYPES OF TESTING

Testing is broadly divided into two main categories:



Functional Testing: Functional Testing checks WHAT the system does based on requirements.

It verifies:

- Features
- User actions
- Outputs
- Business flows
- Data validation

Common Types of Functional Testing

1. Unit Testing

- Testing **individual components or functions** of the application.
- **Example:**
  Testing whether the "calculateTotal()" function returns the correct amount.

2. Integration Testing

- Checks how **two or more modules work together**.

- **Example:**
  Login module → User Dashboard
  Ensuring data flows correctly between modules.

3. System Testing

- Testing the **entire application** end-to-end.
- **Example:**
  E-commerce full workflow:
  Homepage → Product Search → Add to Cart → Payment → Order Confirmation.

4. Acceptance Testing

- Testing done by **end-users or clients** before final release.
- **Example:**
  Client tests:
  "Can I place an order easily using mobile?"

ACTIVITY 4

PRINCIPLES OF TESTING

1. Testing Shows the Presence of Defects

Testing can show that **defects exist**, but it can never prove that **no defects exist**.

- Testing reduces the risk of defects.
- Even if all tests pass, the software may still contain hidden bugs.

Example:
All login test cases pass, but an uncommon browser might still cause an issue.

2. Exhaustive Testing is Impossible

You *cannot* test everything (all combinations, inputs, browsers, devices).

Meaning:

- Testing every input and path is unrealistic.
- Testers focus on **important** and **high-risk** areas.

Example:
Testing all possible password combinations is impossible.

3. Early Testing Saves Time and Money

The earlier you start testing, the cheaper it is to fix defects.

Meaning:

- Testing starts during requirements and design phases.
- Finding issues early prevents costly rework.

Example:
Fixing a requirement misunderstanding is cheaper than fixing wrong code later.

4. Defects Cluster Together

Most defects are usually found in **a few high-risk areas**.

Meaning:

8 of 11

- Some modules have more bugs due to complexity.
- Testers spend more time on high-error-prone areas.

Example:
Payment module usually has more bugs than a homepage.

5. Pesticide Paradox

Meaning :

Using the same test cases again and again will stop finding new bugs.

Example :

If you always test login with the same inputs (correct/wrong password),
you will stop finding new issues — but trying new inputs like long passwords or specia
characters may reveal new bugs.

6. Testing is Context-Dependent

Meaning :

Testing changes based on the type of application.

Example :

A banking app needs strong security testing, but a game app needs graphics and perfo
testing. Different apps → different testing methods.

7. Absence of Error Fallacy

Meaning:

Even if there are no bugs, the software can still fail if it doesn't meet user needs.

Example

An app works perfectly but does NOT have the features the customer asked for —
so the user is still unhappy, and the product fails.

ACTIVITY 5

BRD AND FS FOR AMAZON

BRD -Business Requirement Document

Req_1. User Registration: The system should allow users to create an account using email/phone number.

Req_2. User Login & Authentication: Users should be able to securely log in using pa or OTP.

Req_3. Profile Management: Users should update personal details like name, addres phone number.

Req4. Product Browsing: Users should browse items by category, brand, price, or off

Req_5. Product Search: Users should search products using keywords, filters, and s options.

Req_6. Product Detail Page: Users should view complete product information: imag price, description, reviews.

Req_7. Product Recommendations: The system should display recommended item on user history.

Req_8. Add to Cart: Users should add, update, or remove items in the shopping cart

Req_9. Wishlist: Users should save products in a wishlist for future interest.

Req_10. Checkout Process: System should support smooth checkout with address selection and payment method.

Req_11. Multiple Payment Methods: Users should pay via card, UPI, net banking, wa Cash on Delivery.

Req_12. Order Confirmation: After payment, users should receive order confirmatio email/SMS.

Req_13. Order Tracking: Users should check live order status from packed → shippe delivered.

Req_14. Order Cancellation: Users should cancel orders before shipment or reques delivery return.

Req_15. Return & Refund Process: Users should request returns and track refund st

Req_16. Customer Reviews & Ratings: Users should rate products and write review purchase.

Req_17. Seller Dashboard: Sellers should upload items, update stock, and track thei

Req_18. Inventory Management: System should automatically update stock after ea purchase.

Req_19. Customer Support: Users should contact support via chat, phone, or email issues.

Req_20. Notifications & Alerts: System should send alerts for promotions, order upc delivery reminders, and offers.

FS – Functional Specification

Fs_1 User Registration: System must allow users to sign up using email or mobile nu

Fs_2 User Login: System must authenticate users using password or OTP.

FS_3 Product Search: System must return relevant products based on user keywords

FS_4 Product Filters: System must filter products by brand, price, rating, and catego

FS_5 Product Details: System must display complete product information on the prc page.

FS_6 Add to Cart: System must allow users to add, update, or remove items in the ca

FS_7 Checkout Process: System must let users select address and payment metho place an order.

FS_8 Payment Gateway: System must process payments through cards, UPI, net ba COD.

FS_9 Order Tracking: System must show live status of orders (packed, shipped, deli

FS_10 Review & Ratings: System must allow verified customers to submit product and ratings.